

ORBITAL OVERLAPPING

JEE Advanced Preparation - Chemistry

Overlap Integral, Bonding/Antibonding MOs, Sigma, Pi, Delta

Topic Overview

Orbital overlapping describes how atomic orbitals combine to form molecular orbitals. The sign (phase) of orbitals determines whether overlap produces a bonding (lower energy) or antibonding (higher energy) molecular orbital. This is foundational to understanding chemical bonding.

1. THE OVERLAP INTEGRAL

Overlap Integral: $S = \text{integral of } (\psi_A \times \psi_B) dV$

- ψ_A and ψ_B are the wave functions (orbitals) of atoms A and B
- S is a dimensionless number measuring the extent of orbital overlap
- $S > 0$: Positive overlap $\rightarrow$ constructive interference $\rightarrow$ BONDING MO
- $S < 0$: Negative overlap $\rightarrow$ destructive interference $\rightarrow$ ANTIBONDING MO
- $S = 0$: Zero overlap $\rightarrow$ orbitals are orthogonal $\rightarrow$ no interaction

Condition for bond formation: S must be > 0 (same-phase overlap along internuclear axis)

2. BONDING vs ANTIBONDING MOLECULAR ORBITALS

Bonding MO ($S > 0$):

- Formed when orbitals overlap with SAME phase (constructive interference)
- Electron density INCREASES between the nuclei
- Lower energy than parent atomic orbitals
- Stabilises the molecule
- Notation: σ , π (no star)

Antibonding MO ($S < 0$):

- Formed when orbitals overlap with OPPOSITE phase (destructive interference)
- Electron density DECREASES between the nuclei — has a NODE
- Higher energy than parent atomic orbitals
- Destabilises the molecule
- Notation: σ^* (σ -star), π^* (π -star)

3. PHASE CONVENTION

Atomic orbitals have signs (+/-) representing the **phase of the wave function**. Lobes of the same sign/colour are "in phase"; opposite signs are "out of phase".

Phase Rules:

- Same phase overlap (+/+ or -/-) → constructive → bonding ($S > 0$)
- Opposite phase overlap (+/-) → destructive → antibonding ($S < 0$)
- Orthogonal orientation → zero overlap ($S = 0$) regardless of phase
- Phase is sometimes shown by colour shading: same colour = same phase

4. TYPES OF ORBITAL OVERLAP

4.1 s-s Overlap

s orbitals are spherically symmetric. Two s orbitals ALWAYS overlap positively when they approach along any direction (both are fully symmetric, always in phase with themselves). Gives sigma bonding MO. Example: H-H bond in H_2 .

4.2 s-p Overlap

- Along the axis (head-on): s orbital overlaps with the LARGE positive lobe of p orbital → $S > 0$ → sigma bonding MO (e.g., H-F, H-Cl)
- Perpendicular to p orbital axis: s orbital is equidistant from both + and - lobes of p orbital → equal positive and negative contributions → $S = 0$ (zero, no bond)

4.3 p-p Overlap

Head-on (along axis):

- Same phase (+/+): constructive overlap → sigma BONDING MO ($S > 0$)
- Opposite phase (+/-): destructive overlap → sigma* ANTIBONDING MO ($S < 0$)

Lateral (perpendicular to axis):

- Same phase (both + lobes above axis overlap): pi BONDING MO ($S > 0$)
- Opposite phase (+ lobe of one, - lobe of other overlap): pi* ANTIBONDING MO ($S < 0$)

s-p perpendicular overlap:

- s orbital approaching from the side (perpendicular to p axis): $S = 0$ always

4.4 d Orbital Overlaps — Delta (delta) Bonds

d orbitals can form three types of overlap: sigma (head-on along axis), pi (lateral), and **delta bonds** (face-to-face overlap of d orbitals, e.g., d_{xy} with d_{xy}). Delta bonds occur in metal-metal multiple bonds (e.g., Mo_2 , $\text{Re}_2\text{Cl}_8^{2-}$). delta bonds are weaker than pi bonds due to less effective overlap.

5. ANTIBONDING MO PROPERTIES

Properties of Antibonding MOs:

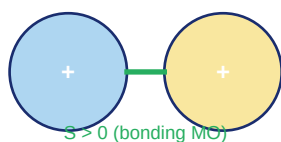
- Have a NODE between the nuclei (region of zero electron density)
- Higher energy than corresponding bonding MO AND parent AOs
- Denoted with asterisk: σ^* , π^* (read sigma-star, pi-star)
- Electrons in antibonding MOs destabilise the bond
- Bond order = (bonding e- minus antibonding e-) / 2
- If bond order = 0, no bond exists (e.g., He_2 would have $\text{BO}=0$)
- π^* MOs are crucial in understanding elimination reactions and conjugation

6. OVERLAP TYPE SUMMARY TABLE

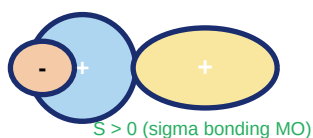
Overlap Type	Phase	S value	MO Formed	Bond Type
s-s head-on	Same (+/+)	$S > 0$	Bonding sigma	Sigma (sigma)
s-p along axis	Same (+/+)	$S > 0$	Bonding sigma	Sigma (sigma)
s-p perpendicular	Any	$S = 0$	No bond	None
p-p head-on	Same (+/+)	$S > 0$	Bonding sigma	Sigma (sigma)
p-p head-on	Opposite (+/-)	$S < 0$	Antibonding sigma*	Sigma* (sigma-star)
p-p lateral	Same (+/+)	$S > 0$	Bonding pi	Pi (pi)
p-p lateral	Opposite (+/-)	$S < 0$	Antibonding pi*	Pi* (pi-star)
d-d face-to-face	Same	$S > 0$	Bonding delta	Delta (delta)

7. ORBITAL OVERLAP DIAGRAMS

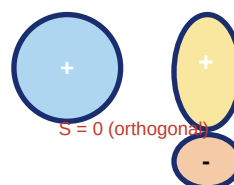
s-s overlap ($S > 0$, bonding)



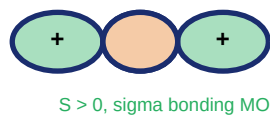
s-p overlap (along axis, $S > 0$)



s-p perpendicular ($S = 0$)



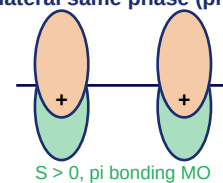
p-p head-on same phase (sigma bond)



p-p opposite phase (sigma* AB)



p-p lateral same phase (pi bond)



Blue = positive lobe (+) | Orange = negative lobe (-) | Green label = bonding | Red label = antibonding

8. WORKED EXAMPLES

Example 1 — Identify overlap type in HCl

Question: What type of orbital overlap forms the H-Cl bond? What is the resulting MO?

Solution:

- H contributes: 1s orbital (spherically symmetric)
- Cl contributes: 3p orbital (directed along the bond axis)
- Overlap type: s orbital of H with the large positive lobe of the 3p orbital of Cl
- This is HEAD-ON (axial) overlap along the internuclear axis
- Both in phase $\rightarrow S > 0 \rightarrow$ BONDING sigma MO
- Answer: s-p head-on overlap; $S > 0$; sigma bonding MO forms the H-Cl bond

Example 2 — Explain why s orbital perpendicular to p orbital gives $S=0$

Question: Show why the overlap integral is zero when an s orbital approaches a p orbital perpendicularly.

Solution:

- p orbital has two lobes: +lobe above axis, -lobe below axis
- When s orbital approaches from the side (perpendicular to p axis):
- The s orbital equally overlaps with BOTH the + lobe and - lobe of p
- Contribution from + lobe: +x (positive overlap)
- Contribution from - lobe: -x (negative overlap)
- Net integral: $+x + (-x) = 0$
- Therefore $S = 0$ by symmetry — no net overlap
- Answer: Symmetry causes equal positive and negative contributions, giving $S = 0$

Example 3 — Count bonding and antibonding electrons in O₂ using MO diagram

Question: Using MO theory, find bond order of O₂ and predict its magnetic nature.

Solution:

- O₂ has 16 electrons total
- MO filling order: $\sigma_{1s}(2)$, $\sigma^*_{1s}(2)$, $\sigma_{2s}(2)$, $\sigma^*_{2s}(2)$, $\sigma_{2p}(2)$, $\pi_{2p}(4)$, $\pi^*_{2p}(2)$
- Bonding electrons: $\sigma_{1s}(2) + \sigma_{2s}(2) + \sigma_{2p}(2) + \pi_{2p}(4) = 10$
- Antibonding electrons: $\sigma^*_{1s}(2) + \sigma^*_{2s}(2) + \pi^*_{2p}(2) = 6$
- Bond order = $(10 - 6)/2 = 4/2 = 2$ (double bond, as expected)
- 2 electrons in degenerate π^*_{2p} MOs are UNPAIRED (Hund rule)
- Unpaired electrons → O₂ is PARAMAGNETIC
- Answer: BO = 2; O₂ is paramagnetic with 2 unpaired electrons in π^* MOs

Example 4 — Identify all types of overlap in ethylene (C₂H₄)

Question: Identify all orbital overlaps in C₂H₄.

Solution:

- C₂H₄: Each C is sp² hybridised
- C-H bonds: s(H) with sp²(C) — s-sp² head-on overlap → sigma bonds (4 such bonds)
- C-C sigma bond: sp²(C) with sp²(C) — head-on overlap → sigma bond
- C=C pi bond: unhybridised p(C) with p(C) — lateral overlap → pi bonding MO
- Total: 5 sigma bonds (s-sp² type) + 1 pi bond (p-p lateral)
- Answer: C₂H₄ has 5 sigma bonds (s-sp²/sp²-sp²) and 1 pi bond (p-p lateral)

Example 5 — Explain why p-p perpendicular overlap is zero

Question: Two p orbitals oriented perpendicular to each other (one along x, one along y). Explain why $S = 0$.

Solution:

- px orbital has lobes along x-axis (+lobe right, -lobe left)
- py orbital has lobes along y-axis (+lobe up, -lobe down)
- These two orbitals are ORTHOGONAL to each other
- The overlap region of px with the +lobe of py is cancelled by the overlap of px with the -lobe of py
- By symmetry: net integral = 0
- This is why orthogonal p orbitals do NOT form bonds with each other
- Answer: Orthogonal orbitals have zero overlap integral by symmetry $\rightarrow S = 0 \rightarrow$ no bond

9. KEY FACTS — QUICK REVISION

KEY FACTS / QUICK REVISION — ORBITAL OVERLAPPING

- S = overlap integral; $S > 0$ bonding, $S < 0$ antibonding, $S = 0$ no interaction
- Same phase overlap $\rightarrow$ constructive interference $\rightarrow$ bonding MO (lower energy)
- Opposite phase $\rightarrow$ destructive interference $\rightarrow$ antibonding MO (higher energy, node)
- s-s: always positive (sigma bond, e.g., H_2)
- s-p along axis: positive sigma bond (e.g., HF, HCl)
- s-p perpendicular: $S = 0$ (by symmetry)
- p-p head-on same phase: sigma bonding MO
- p-p head-on opposite phase: sigma* antibonding MO (node between nuclei)
- p-p lateral same phase: pi bonding MO
- p-p lateral opposite phase: pi* antibonding MO
- Delta bonds: face-to-face d-d overlap (in metal complexes)
- Antibonding MOs starred: sigma*, pi*; always have a nodal plane between atoms
- Bond order = (bonding e- - antibonding e-) / 2
- O_2 is paramagnetic: 2 unpaired e- in degenerate pi*2p antibonding MOs

10. PRACTICE QUESTIONS — JEE ADVANCED LEVEL

1. An orbital overlap with $S > 0$ results in:

- (A) Antibonding MO
- (B) Bonding MO
- (C) Non-bonding MO
- (D) No interaction

2. Why is the overlap of an s orbital with a p orbital perpendicular to its axis equal to zero?

- (A) s orbital is too small
- (B) Equal positive and negative contributions cancel by symmetry
- (C) p orbital has no phase
- (D) s and p orbitals never overlap

3. Which overlap gives a pi bonding MO?

- (A) s-s head-on
- (B) p-p head-on same phase
- (C) p-p lateral same phase
- (D) s-p along axis

4. An antibonding MO differs from a bonding MO in that it:

- (A) Has lower energy
- (B) Has increased electron density between nuclei
- (C) Has a node between the nuclei
- (D) Is formed by constructive interference

5. Bond order of O₂ (using MO theory) is:

- (A) 1
- (B) 2
- (C) 3
- (D) 1.5

6. Which of the following is an example of s-p sigma overlap?

- (A) H-H in H₂
- (B) H-F in HF
- (C) F-F in F₂
- (D) pi bond in O₂

7. The overlap integral S for two orthogonal orbitals is:

- (A) +1
- (B) -1
- (C) 0
- (D) Infinity

8. Which molecular orbital has a nodal plane perpendicular to the internuclear axis?

- (A) sigma bonding MO
- (B) sigma* antibonding MO
- (C) pi bonding MO
- (D) Both sigma and pi bonding MOs

9. Delta (delta) bonds are formed by:

- (A) s-s head-on overlap
- (B) p-p lateral overlap
- (C) d-d face-to-face overlap
- (D) p-p head-on overlap

10. Paramagnetism of O₂ is explained by MO theory because:

- (A) O₂ has zero bond order
- (B) 2 electrons are unpaired in degenerate π^* antibonding MOs
- (C) All electrons in O₂ are in bonding MOs
- (D) O₂ has σ^* electrons only

11. ANSWER KEY WITH EXPLANATIONS

1. Answer: B — Bonding MO

$S > 0$ means same-phase constructive interference. This produces a bonding MO with increased electron density between nuclei and lower energy than parent AOs.

2. Answer: B

When s approaches p perpendicularly, it equally overlaps with the +lobe and -lobe of the p orbital. These contributions cancel exactly: $\text{integral}(s \times p+) + \text{integral}(s \times p-) = +k + (-k) = 0$.

3. Answer: C — p-p lateral same phase

π bonds are formed by lateral (sideways) overlap of p orbitals. Same phase means constructive interference $\rightarrow$ π BONDING MO. Head-on p - p gives sigma; s - s and s - p give sigma.

4. Answer: C — Has a node between the nuclei

Antibonding MOs have destructive interference, creating a nodal plane between the nuclei where electron density = 0. They have HIGHER energy (not lower) than parent AOs.

5. Answer: B — Bond order 2

O_2 MO: bonding $e^- = 10$, antibonding $e^- = 6$. $BO = (10-6)/2 = 2$. This is a double bond, consistent with $O=O$ structure. The 2 unpaired electrons in π^*2p make it paramagnetic.

6. Answer: B — H-F in HF

H contributes $1s$ orbital; F contributes $2p$ orbital. Along the bond axis, this is s - p head-on overlap $\rightarrow$ sigma bond. H-H is s - s ; F-F is p - p ; π in O_2 is p - p lateral.

7. Answer: C — 0

Orthogonal orbitals have zero overlap by symmetry. The positive contributions from one lobe cancel the negative contributions from the other lobe.

8. Answer: B — sigma* antibonding MO

The σ^* (sigma antibonding) MO has a nodal PLANE perpendicular to the internuclear axis, passing through the midpoint between the nuclei. This is the hallmark of antibonding MOs.

9. Answer: C — d-d face-to-face overlap

Delta bonds form when d orbitals overlap face-to-face (e.g., dxy with dxy). They occur in transition metal multiple bonds. Bond strength: $\sigma > \pi > \delta$.

10. Answer: B

O₂ has 12 valence electrons in MOs. After filling σ_{2p} and π_{2p} MOs, 2 electrons go into the two degenerate π^*_{2p} MOs — one each (Hund's rule) → 2 unpaired electrons → paramagnetic. VBT cannot explain this.